

Regulation of In Vitro Palmitate Oxidation in Liver from Dairy Cows During Early Lactation¹

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ABSTRACT

Regulatory effects of carnitine, glucose, some glucogenic compounds (propionate, pyruvate, alanine, lactate, glycerol, and fructose), ketone bodies (acetate, acetoacetate, and β -hydroxybutyrate), and insulin on oxidation of palmitate were studied in slices of liver obtained from high producing dairy cows during early lactation. A total of 77 biopsies of liver from 21 multiparous Holstein cows (36 ± 16 d postpartum) was used. L-Carnitine increased oxidation of palmitate to CO_2 by more than twofold and oxidation to acid-soluble products by about fourfold. Propionate decreased oxidation of palmitate in liver slices incubated without added carnitine, but the decrease was lessened by carnitine. Pyruvate, lactate, and alanine increased palmitate oxidation, especially in the presence of carnitine. Glycerol, glucose, and insulin tended to decrease palmitate oxidation in the absence of carnitine. Fructose tended to decrease oxidation to CO_2

but did not affect oxidation to acid-soluble products. Acetate and acetoacetate decreased oxidation of palmitate, whereas β -hydroxybutyrate decreased palmitate oxidation in the absence of carnitine but increased palmitate oxidation in its presence. In general, carnitine decreased the inhibitory effects of compounds that decreased palmitate oxidation but increased the stimulatory effects of compounds that increased palmitate oxidation.

(Key words: liver, fatty acids, lactating cow)

Abbreviation key: ASP = acid-soluble products, BHBA = β -hydroxybutyrate, CoA = coenzyme A, CPT-I = carnitine palmitoyltransferase I, ISU = Iowa State University, UI = University of Illinois.

INTRODUCTION

High producing dairy cows respond to negative energy balance during early lactation by mobilizing large quantities of NEFA from adipose tissue. The liver takes up NEFA in proportion to their concentration in blood (23), and NEFA within liver are either esterified or oxidized to CO_2 or ketone bodies. Excessive uptake of NEFA relative to availability of gluconeogenic precursors during early lactation can lead to metabolic disorders such as ketosis and fatty liver (2). To understand the etiology of these disorders, therefore, it is important to identify factors that regulate oxidation of NEFA by liver in dairy cows during early lactation.

Control of hepatic metabolism of NEFA is understood poorly in dairy cows. Oxidative

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capacity, as measured by *in vitro* oxidation of palmitate and total carnitine palmitoyltransferase I (CPT-I; EC 2.3.1.21) activity, was greatest in cows during early lactation (1). Regulation of oxidation via modulation of CPT-I activity, however, is thought to be more important in ruminants than are changes in total CPT-I activity (7). Metabolites or hormones that may modulate activity of CPT-I or affect other steps of NEFA oxidation can be identified by using isolated tissue preparations. Regulatory mechanisms then can be studied in more detail in cell or tissue culture systems or in the whole animal.

Control of hepatic oxidation of NEFA in ruminants seems to be linked to metabolism of carbohydrates as in nonruminants. Propionate and other glucogenic compounds inhibited oxidation of NEFA by bovine liver slices (1, 13, 14) and ovine hepatocytes (16). Metabolic intermediates such as malonyl-coenzyme A (CoA) and methylmalonyl-CoA also are potent inhibitors of CPT-I activity in sheep hepatocytes (5). It has been suggested that the ketone bodies [i.e., acetoacetate and β -hydroxybutyrate (BHBA)] may autoregulate their own production in ruminant liver (12), but this suggestion has not been verified *in vitro*.

Evidence suggests that regulation of NEFA oxidation by carbohydrate intermediates may be dependent upon carnitine content of the liver. Carnitine increased oxidation of NEFA in bovine liver slices (13) and ovine hepatocytes (16). Supplementation of carnitine to ovine hepatocytes decreased the antiketogenic potential of several glucogenic compounds added to the media (16). Further, exogenous carnitine did not stimulate oxidation of NEFA by hepatocytes from fasted sheep (16) in which hepatic carnitine concentrations are increased greatly relative to fed sheep (21). Fasting also decreased the ability of glucogenic compounds to inhibit oxidation of NEFA (16). These observations suggest that interactions between carnitine and other metabolic compounds could be important in regulation of NEFA oxidation in dairy cows during early lactation, when mobilization of NEFA and hepatic ketogenesis are major processes.

The objective of this experiment was to determine regulatory effects of carnitine, several glucogenic compounds, insulin, and ketone bodies during *in vitro* oxidation of palmitate

by liver slices obtained from dairy cows during early lactation. Specific aims were 1) to extend observations of Jesse et al. (13, 14) to high producing cows in early lactation, 2) to test effects of additional glucogenic compounds not used in earlier studies (13, 14), 3) to test the hypothesis that ketone bodies decrease NEFA oxidation in bovine liver, and 4) to measure possible differences in response to metabolic effectors caused by addition of carnitine to the incubation media.

MATERIALS AND METHODS

Materials

L-Carnitine was a gift from Lonza, Inc. (Fair Lawn, NJ). [1- 14 C]Palmitic acid was purchased from ICN Biomedicals, Inc. (Costa Mesa, CA). Sodium palmitate, fatty acid-free bovine serum albumin, L-alanine, α -D-glucose, glycerol, β -D-fructose, sodium salts of propionate, pyruvate, D-BHBA, and acetate, lithium salts of L-lactate and acetoacetate, and bovine insulin were purchased from Sigma Chemical Company (St. Louis, MO). Scintillation cocktail (Ready-Solv EP[®]) was from Beckman Instruments, Inc. (Fullerton, CA). All chemicals were of reagent or higher grade.

Methods

Liver samples were obtained from a total of 77 biopsies from 21 multiparous Holstein cows. Cows averaged 36 ± 16 d postpartum (range 12 to 67) and produced 33.5 kg milk/d (range 23 to 47, SD 4.9) at the time of liver biopsy. Twelve cows were from a study at Iowa State University (ISU) designed to induce ketonemia through moderate feed restriction and dietary 1,3-butanediol (10). These cows were fed a diet (DM basis) of 31% corn silage, 5% alfalfa haylage, 7% alfalfa hay, and 57% concentrate (corn, soybean meal, vitamins, minerals). Feeding and management of the ISU cows are described in detail elsewhere (10). From four to six biopsies were taken from each of the ISU cows at least 1 wk apart. Nine cows in the University of Illinois (UI) herd were biopsied once each. These cows were fed a total mixed diet once daily that consisted of 25% alfalfa haylage, 25% corn silage, and 50% concentrate (corn, soybean meal, vitamins, and minerals).

Liver samples were obtained by puncture biopsy under local anesthesia (24) and placed immediately into ice-cold phosphate-buffered (.015 M) .9% NaCl at pH 7.4. Tissue was used for metabolic assays within 1 h. Liver was sliced by hand on a Stadie-Riggs microtome to .5-mm thickness (70 to 130 mg). Slices were used, despite their inherently low rates of palmitate oxidation, because of the difficulties experienced by other investigators in isolating sufficient numbers of viable hepatocytes from bovine liver obtained by core biopsy (R. J. Aiello and L. E. Armentano, University of Wisconsin, and A. Gronowski and A. Trenkle, ISU, personal communications). Slices were blotted, weighed, and placed into 25-ml flasks that contained 3 ml of Ca-free Krebs-Henseleit bicarbonate buffer (15) at pH 7.4. Substrate was sodium palmitate (2 mM) with .3 μ Ci of [1^{14} C]palmitic acid complexed to fatty acid-free bovine serum albumin in a 4:1 molar ratio. Effects of the glucogenic compounds, ketone bodies, and insulin listed under Materials were measured both with and without supplemental L-carnitine (1 mM).

A hanging well containing a folded filter paper was inserted into each flask. Flasks then were gassed with 95:5 O₂:CO₂, sealed with a rubber septum cap, and incubated for 2 h with constant shaking in a water bath at 37°C. Incubations were terminated by injecting 3 ml of 10% (wt/vol) perchloric acid, and .1 ml of 30% NaOH was injected onto the suspended filter paper. Flasks were shaken for an additional 1 h to collect evolved CO₂. Filter papers then were removed, the center well was rinsed, and filter paper and rinses were added to a scintillation vial. The liquid was evaporated, 10 ml of scintillation cocktail were added, and the vials were counted in a scintillation counter to measure oxidation of palmitate to CO₂. Rates of oxidation of palmitate to CO₂ were linear for up to 3 h, both in the presence and absence of carnitine.

Flasks were allowed to cool to room temperature, and the medium was poured into 50-ml polyethylene centrifuge tubes, which then were centrifuged for 10 min at 2000 $\times$ g. Two milliliters of the supernatant were transferred to a 15 $\times$ 85-mm test tube and neutralized to pH 7 with 2 M K₂CO₃. After centrifuging for 10 min at 2500 $\times$ g, .5 ml of the final supernatant was transferred to a scintillation

vial, 10 ml of scintillation cocktail were added, and the vials were counted to measure incomplete oxidation of palmitate to acid-soluble products (ASP). In studies similar to ours, ketone bodies and acetate accounted for about 80% of the radiolabeled ASP, with citric acid cycle and glycolytic intermediates making up most of the rest (13). Measuring radioactivity in both ASP and CO₂ provides a more complete estimate of total oxidation of NEFA than does measuring radioactive CO₂ alone (25). Similar to results of Jesse et al. (13), rates of oxidation of palmitate to ASP were greater during the first 15 min of incubation but then were linear through 3 h. Rates were calculated on the basis of total oxidation to ASP during 2 h of incubation.

Because of limitations on the amount of liver that could be obtained safely from each biopsy and the number of flasks that could be assayed on any one day, not all compounds could be tested for each liver biopsy. Within each liver biopsy, compounds whose effects on palmitate oxidation were to be tested were compared with appropriate control flasks that contained only palmitate or palmitate plus carnitine. Triplicate flasks were incubated for each comparison, except for occasional instances in which the amount of liver obtained allowed only duplicate flasks. Average rates of replicate flasks were corrected for background radioactivity by subtracting values obtained for flasks acidified immediately after addition of liver slices. Data are expressed as nanomoles of palmitate oxidized to product (i.e., CO₂ or ASP) per hour per gram of liver (wet weight) because variation in lipid content of the liver was minimal (10).

Responses to all compounds tested were not correlated ($P > .15$) with location (ISU or UI), day of lactation, milk production, or energy intake. For ISU cows, therefore, all repeated measurements of effects of a compound within a cow were averaged to provide a single value for that cow. Statistical comparisons then were made by using paired t tests (22). Significance was ($P < .05$).

RESULTS AND DISCUSSION

Rates of oxidation of palmitate to CO₂ and ASP by bovine liver slices in the absence of added carnitine (Table 1) generally were

TABLE 1. Effects of L-carnitine (1 mM) on oxidation of palmitate by bovine liver slices to carbon dioxide and total acid-soluble products (ASP).¹

Media	Product			
	CO ₂		ASP	
	(nmol/h per g tissue)			
	$\bar{X}$	SE	$\bar{X}$	SE
Without carnitine	21.4	1.6	161.0	10.1
With carnitine	45.8*	4.2	637.1*	28.1

¹n = 18.*Different from paired incubations without carnitine ($P < .0001$).

greater than reported by others (13, 14), which may reflect the earlier stage of lactation of our cows. Addition of L-carnitine to incubation media increased oxidation of palmitate to CO₂ by more than twofold and to ASP by almost fourfold (Table 1). The magnitude of stimulation by carnitine was similar to that of previous reports (13). Actual carnitine-stimulated rates of oxidation to CO₂ were somewhat less, and rates of oxidation to ASP greater, than reported previously (13, 14).

Propionate decreased oxidation of palmitate to CO₂ by more than half without supplement-

tal carnitine but only by about one-fifth with carnitine (Table 2). Others showed that propionate decreased oxidation of palmitate to CO₂ in bovine liver slices (13) but increased oxidation of palmitate to CO₂ by isolated ovine hepatocytes incubated without carnitine (16). Other glucogenic compounds or insulin did not significantly affect oxidation of palmitate to CO₂ in the absence of exogenous carnitine (Table 2). Others found that lactate, fructose, and pyruvate increased oxidation to CO₂ by ovine hepatocytes (16).

In the presence of carnitine (Table 2), pyruvate and alanine increased oxidation to CO₂, whereas glycerol (5 mM), fructose, and glucose (10 mM) tended to decrease oxidation to CO₂. Studies by others in which bovine liver slices were incubated with added carnitine showed that glucose and insulin decreased oxidation of palmitate to CO₂ (14), whereas lactate, fructose, and pyruvate increased oxidation to CO₂ in ovine hepatocytes (16).

Propionate decreased oxidation of palmitate to ASP without exogenous carnitine in the medium, but responses were variable and not significant with added carnitine (Table 3). Similar findings were reported with ovine hepatocytes incubated with carnitine and propionate (16). Pyruvate increased oxidation to

TABLE 2. Mean change in oxidation of palmitate to carbon dioxide by bovine liver slices incubated with glucogenic compounds, glucose, or insulin either with or without added L-carnitine.

Addition ¹	Without carnitine				With carnitine (1 mM) ¹			
	Mean change ²	SE ³	n ⁴	P ⁵	Mean change ²	SE ³	n ⁴	P ⁵
	(nmol/h per g tissue)							
Propionate (5 mM)	-12.6	1.3	7	<.0001	-8.9	2.6	10	.008
Pyruvate (5 mM)	1.8	2.5	9	.50	20.9	5.2	10	.004
Lactate (5 mM)	.6	1.7	8	.73	1.5	3.6	8	.69
Alanine (5 mM)	-.1	1.2	8	.92	8.6	2.5	8	.01
Glycerol (.5 mM)	-.1	1.0	8	.89	...	...	...	...
Glycerol (5 mM)	-.3	1.1	7	.78	-9.0	3.3	2	.10
Fructose (5 mM)	-4.1	2.9	8	.20	-3.0	1.4	8	.08
Glucose (5 mM)	-1.5	1.8	8	.42	...	...	...	...
Glucose (10 mM)	-.5	2.1	6	.83	-4.9	3.1	10	.14
Insulin (780 nM)	.7	1.0	6	.52	-2	2.7	10	.97

¹Concentrations of additions are in parentheses.²Change in rate of oxidation from paired controls containing only palmitate or palmitate plus carnitine as appropriate. Average rates of controls were similar to those in Table 1.³Standard error of the difference in rates from paired controls.⁴Number of cows in which effects were tested.⁵Probability of difference from paired controls.

TABLE 3. Mean change in oxidation of palmitate to total acid-soluble products (ASP) by bovine liver slices incubated with glucogenic compounds, glucose, or insulin either with or without added L-carnitine.

Addition ¹	Without carnitine				With carnitine (1 mM) ¹			
	Mean change ²	SE ³	n ⁴	P ⁵	Mean change ²	SE ³	n ⁴	P ⁵
	(mmol/h per g tissue)							
Propionate (5 mM)	-50.9	6.4	7	.0003	-3.7	36.8	10	.92
Pyruvate (5 mM)	21.0	7.1	9	.02	192.1	47.6	10	.003
Lactate (5 mM)	7.2	5.0	8	.20	61.2	20.4	8	.02
Alanine (5 mM)	29.0	9.4	8	.02	209.6	42.2	8	.002
Glycerol (5 mM)	-14.9	15.8	8	.38	...	...	...	...
Glycerol (5 mM)	-23.8	5.0	7	.004	-56.9	42.1	2	.41
Fructose (5 mM)	.5	6.2	8	.94	-1.9	33.7	8	.96
Glucose (5 mM)	-19.3	11.9	8	.15	...	...	...	...
Glucose (10 mM)	-14.3	10.9	6	.25	-17.3	33.6	10	.62
Insulin (780 nM)	-21.3	9.5	6	.08	-33.5	29.1	10	.28

¹Concentrations of additions are in parentheses.²Change in rate of oxidation from paired controls containing only palmitate or palmitate plus carnitine as appropriate. Average rates of controls were similar to those in Table 1.³Standard error of the difference in rates from paired controls.⁴Number of cows in which effects were tested.⁵Probability of difference from paired controls.

ASP both with and without exogenous carnitine, although the effect was much greater in the presence of carnitine (Table 3). Alanine also increased palmitate oxidation to ASP both with and without added carnitine (Table 3). Lactate significantly increased oxidation of palmitate to ASP only in the presence of exogenous carnitine (Table 3).

In high concentrations (5 mM), glycerol decreased oxidation to ASP in incubations without added carnitine (Table 3). Antiketogenic effects of glycerol also were observed in starved sheep (16). Effects of fructose were highly variable and nonsignificant (Table 3), in contrast with other studies in which fructose decreased oxidation of palmitate in sheep hepatocytes (16).

Glucose added at concentrations of either 5 or 10 mM did not significantly decrease oxidation to ASP either with or without added carnitine (Table 3). Insulin tended ($P < .08$) to decrease oxidation in incubations without carnitine (Table 3), but the response was more variable with added carnitine. Other researchers reported that glucose and insulin in the presence of carnitine decreased oxidation of palmitate to both CO₂ and ASP by bovine liver slices (14). Responses to glucose and insulin were less than in other reports (14), which would be consistent with the greater

resistance to actions of insulin during early lactation.

Oxidation of palmitate to CO₂ was decreased by BHBA without exogenous carnitine but tended ($P < .12$) to be increased when carnitine was added to the medium (Table 4). Acetoacetate also tended ($P < .14$) to increase oxidation to CO₂ in the presence of carnitine (Table 4). Acetate decreased oxidation to CO₂ both with and without carnitine (Table 4), which is in agreement with findings of Jesse et al. (14).

In incubations without added carnitine, BHBA decreased oxidation of palmitate to ASP but increased oxidation to ASP in the presence of carnitine (Table 5). Acetoacetate significantly decreased oxidation to ASP in the absence of carnitine, but significance of the decrease was lost in the presence of carnitine because of increased variation (Table 5). Acetate tended ($P < .11$) to decrease oxidation of palmitate to ASP in the absence of carnitine and significantly decreased oxidation to ASP in the presence of carnitine (Table 5), which is consistent with other results (14).

CONCLUSIONS

Our results provide an extension of studies of effects of several metabolic intermediates on

TABLE 4. Mean change in oxidation of palmitate to carbon dioxide by bovine liver slices incubated with ketone bodies either with or without added L-carnitine.

Addition ¹	Without carnitine				With carnitine (1 mM) ¹			
	Mean change ²	SE ³	n ⁴	P ⁵	Mean change ²	SE ³	n ⁴	P ⁵
(nmol/h per g tissue)								
BHBA ⁶ (5 mM)	-2.4	.8	8	.02	...	...	...	...
BHBA (10 mM)	-2.7	1.0	7	.004	2.6	1.4	8	.12
Acetoacetate (5 mM)	.8	1.6	8	.62	5.8	3.5	8	.14
Acetate (5 mM)	-2.9	.3	5	.0009	-4.6	.8	5	.005

¹Concentrations of additions are in parentheses.²Change in rate of oxidation from paired controls containing only palmitate or palmitate plus carnitine as appropriate. Average rates of controls were similar to those in Table 1.³Standard error of the difference in rates from paired controls.⁴Numbers of cows in which effects were tested.⁵Probability of difference from paired controls.⁶D-β-Hydroxybutyrate.

oxidation of palmitate in liver slices from dairy cows. We have extended the limited number of compounds already tested in bovine liver slices (13, 14). A major strength of our data is that effects of these compounds were measured in liver from cows during early lactation; thus, our results can be applied more directly to high producing cows. Responses of liver from our high producing cows during early lactation to many of the compounds, however, were qualitatively similar to results of previous

work with cows in later lactation or in nonlactating ruminants (13, 14, 16). Discussion will center on aspects of regulation unique to our study or that differ qualitatively from previous reports.

The dramatic stimulation of palmitate oxidation by carnitine suggests that a lack of carnitine could limit maximal rates of palmitate oxidation in lactating cows. Carnitine recently was suggested to be more important in regulation of NEFA oxidation in ruminants

TABLE 5. Mean change in oxidation of palmitate to total acid-soluble products (ASP) by bovine liver slices incubated with ketone bodies either with or without added L-carnitine.

Addition ¹	Without carnitine				With carnitine (1 mM) ¹			
	Mean change ²	SE ³	n ⁴	P ⁵	Mean change ²	SE ³	n ⁴	P ⁵
(nmol/h per g tissue)								
BHBA ⁶ (5 mM)	-14.6	7.8	8	.11	...	...	...	...
BHBA (10 mM)	-30.7	6.4	7	.004	52.5	18.7	8	.03
Acetoacetate (5 mM)	-31.1	11.0	8	.02	-30.1	25.6	8	.28
Acetate (5 mM)	-36.8	18.1	5	.11	-81.7	28.1	5	.04

¹Concentrations of additions are in parentheses.²Change in rate of oxidation from paired controls containing only palmitate or palmitate plus carnitine as appropriate. Average rates of controls were similar to those in Table 1.³Standard error of the difference in rates from paired controls.⁴Numbers of cows in which effects were tested.⁵Probability of difference from paired controls.⁶D-β-Hydroxybutyrate.

than previously thought in nonruminants (7). Little is known about carnitine metabolism in dairy cows, although carnitine content of liver is greater during lactation than during the non-lactating state (21). Carnitine concentration increases dramatically in liver of sheep during fasting (21). During fasting, breakdown of protein in skeletal muscle releases trimethyllysine, which is converted to carnitine in the liver (9). If hepatic carnitine concentration increases during the catabolic conditions of negative energy balance during early lactation, our results suggest that carnitine could be important in regulating ketogenesis from NEFA.

Alternatively, stimulation of palmitate oxidation by carnitine could be an effect observed only in isolated tissues. Carnitine is lost from hepatocytes during isolation procedures (8). Although content of carnitine in our liver slices was not measured, similar losses of carnitine may have occurred during slicing of liver. Further study of factors affecting concentrations of carnitine in bovine liver, both in vitro and in vivo, is warranted.

The inhibitory effect of propionate on fatty acid oxidation and ketogenesis is well documented (6, 13, 16) although the mechanism for the effect remains unclear. Jesse et al. (13) hypothesized that propionate decreased palmitate oxidation by increasing its esterification. Others have proposed both intramitochondrial (6, 16) and cytosolic (5) mechanisms whereby propionate decreases oxidation of fatty acids. That carnitine seemed to decrease the inhibitory effects of propionate agrees with previous work with rats (4) and sheep (16); this effect of carnitine has been attributed to increased formation of propionylcarnitine (4). Although we made no measurements of carnitine esters, propionate was added to the medium at five times the concentration of carnitine; it seems unlikely, therefore, that formation of propionylcarnitine is the sole mechanism whereby carnitine decreases the inhibitory effects of propionate on palmitate oxidation.

Most of the other glucogenic compounds had rather modest effects on oxidation of palmitate, the notable exceptions being pyruvate and alanine. Stimulation of palmitate oxidation by pyruvate could be the result of an increased need for reducing equivalents and ATP to synthesize glucose from pyruvate (11); however, no measure of glucose production was made.

Reasons for the greater stimulation by pyruvate in the presence of carnitine are unclear. Explanations for the increased palmitate oxidation in the presence of lactate probably involve demands for ATP for gluconeogenesis, because gluconeogenesis from lactate is thought to be balanced in terms of reducing equivalents (11).

Alanine produced increases of palmitate oxidation similar to the increases caused by pyruvate. Alanine is antiketogenic in rats (18), but we measured total oxidation of palmitate rather than production of ketone bodies. The increase in oxidation of palmitate probably was needed to supply ATP for gluconeogenesis and urea synthesis. Gluconeogenesis from alanine, coupled with synthesis of urea for disposal of amino groups, is balanced in supply and demand for reducing equivalents (11). Alanine is a major carrier of carbon from peripheral tissues to liver for gluconeogenesis, which may be particularly important during the negative energy balance of early lactation. Increased supply of alanine to the liver, therefore, could help to signal negative glucose balance and the need for fatty acid oxidation to supply energy for gluconeogenesis.

Ketone bodies decrease ketogenesis in vivo, both by direct inhibition of lipolysis and by stimulation of insulin release, which in turn would suppress lipolysis (20). Our data tend to support the hypothesis that an additional site of regulation of NEFA oxidation by ketone bodies may be within the liver itself (12). Acetoacetate seemed to suppress oxidation of palmitate to ASP but to increase oxidation to CO_2 . This implies that autoregulation of ketogenesis may not be at the level of entry of palmitate into the mitochondria via CPT-I. If similar effects occur in vivo, increased acetoacetate concentrations during ketosis could act to decrease ketogenesis and increase activity of the citric acid cycle and, therefore, help in self-alleviation of ketosis.

The differential effect of BHBA on palmitate oxidation, which is dependent upon carnitine status, carries different connotations for development of ketosis. If hepatic concentrations of carnitine increase during negative energy balance (21), increased production of BHBA from NEFA could tend to stimulate rather than inhibit ketogenesis and thereby aggravate ketotic conditions. Thus, the interac-

tions of the ketone bodies and carnitine on palmitate oxidation deserve further investigation.

As noted by Jesse et al. (14), acetate was a potent inhibitor of palmitate oxidation. Supply of acetate from rumen fermentation, which would be indicative of amount of feed intake and relative energy balance, thus could be linked to the need for fatty acid oxidation. The nature of the inhibition by acetate is unclear. Uptake and oxidation of acetate by ruminant liver is believed to be minimal because of the low activity of acetyl-CoA synthetase [EC 6.2.1.1; (19)].

One possible mechanism for the decreased oxidation of fatty acids by acetate could be through the limited activity of the fatty acid synthesis pathway in ruminant liver (3). Malonyl-CoA is a powerful inhibitor of NEFA oxidation in sheep liver (5). Concentrations of malonyl-CoA in sheep liver were of similar magnitude to those of rat liver and changed with physiological state similar to rats (5). The presence of low activities of the fatty acid synthetic pathway thus could serve a regulatory function in ruminant liver, rather than serving as a quantitatively important source of long-chain fatty acids.

Carnitine generally increased the stimulation of oxidation and decreased the inhibition of oxidation by those compounds that affected palmitate oxidation. Exact mechanisms of these interactions cannot be determined from our study. A major unresolved factor in studies of this type is the identification of the products of metabolism in the ASP fraction. Although the ASP fraction often is used as an indicator of ketogenesis, carnitine esters make up a variable and perhaps major fraction of ASP in some circumstances (17). Carnitine may serve as a CoA buffer by its ability to form carnitine esters (4). The radiolabeled products found in the ASP fraction should be identified in any future experiments designed to study the nature of interactions between carnitine and other metabolites.

The effects and possible roles of carnitine, propionate, alanine, and the ketone bodies in regulating hepatic oxidation of long-chain fatty acids are especially interesting in relation to metabolic disorders during the critical early postpartum period. Future studies should seek to determine the significance of these effects in

cell or tissue culture systems and in vivo. Identification of mechanisms of interaction between carnitine and metabolism of carbohydrates, fatty acids, and amino acids could lead to strategies to decrease incidence and severity of metabolic disorders and increase productive efficiency during early lactation.

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REFERENCES

- 1 Aiello, R. J., T. M. Kenna, and J. H. Herbein. 1984. Hepatic gluconeogenic and ketogenic interrelationships in the lactating cow. *J. Dairy Sci.* 67:1707.
- 2 Baird, G. D. 1982. Primary ketosis in the high producing cow: clinical and subclinical disorders, treatment, prevention, and outlook. *J. Dairy Sci.* 65:1.
- 3 Ballard, F. J., R. W. Hanson, and D. S. Kronfeld. 1969. Gluconeogenesis and lipogenesis in tissue from ruminant and nonruminant animals. *Fed. Proc.* 28:218.
- 4 Brass, E. P., P. V. Fennessey, and L. V. Miller. 1986. Inhibition of oxidative metabolism by propionic acid and its reversal by carnitine in isolated rat hepatocytes. *Biochem. J.* 236:131.
- 5 Brindle, N.P.J., V. A. Zammit, and C. I. Pogson. 1985. Regulation of carnitine palmitoyltransferase activity by malonyl-CoA in mitochondria from sheep liver, a tissue with a low capacity for fatty acid synthesis. *Biochem. J.* 232:177.
- 6 Bush, R. S., and L. P. Milligan. 1971. Study of the mechanism of inhibition of ketogenesis by propionate in bovine liver. *Can. J. Anim. Sci.* 51:121.
- 7 Butler, S. M., A. Faulkner, V. A. Zammit, and R. G. Vernon. 1988. Fatty acid metabolism of the perfused caudate lobe from livers of fed and fasted non-pregnant and fasted late pregnant ewes. *Comp. Biochem. Physiol.* 91B:25.
- 8 Christiansen, R. Z., and J. Bremer. 1976. Active transport of butyrobetaine and carnitine into isolated liver cells. *Biochim. Biophys. Acta* 448:562.
- 9 Davis, A. T., and C. L. Hoppel. 1986. Effect of starvation on the disposition of free and peptide-linked trimethyllysine in the rat. *J. Nutr.* 116:760.
- 10 Drackley, J. K. 1989. Factors contributing to development of fatty liver and ketosis in lactating dairy cows. Ph.D. Diss., Iowa State Univ., Univ. Microfilms, Ann Arbor, MI.
- 11 Harris, R. A. 1986. Carbohydrate metabolism I: major metabolic pathways and their control. Page 261 in *Textbook of biochemistry with clinical correlations*. T. M. Devlin, ed. John Wiley and Sons, New York, NY.
- 12 Heitmann, R. N., D. J. Dawes, and S. C. Sensenig. 1987. Hepatic ketogenesis and peripheral ketone body

- utilization in the ruminant. *J. Nutr.* 117:1174.
- 13 Jesse, B. W., R. S. Emery, and J. W. Thomas. 1986. Control of bovine hepatic fatty acid oxidation. *J. Dairy Sci.* 69:2290.
- 14 Jesse, B. W., R. S. Emery, and J. W. Thomas. 1986. Aspects of the regulation of long-chain fatty acid oxidation in bovine liver. *J. Dairy Sci.* 69:2298.
- 15 Lasser, H. 1961. Salt solutions. Page 58 in *Biochemists handbook*. C. Long, ed. D. Van Nostrand Co., Inc., Princeton, NJ.
- 16 Lomax, M. A., I. A. Donaldson, and C. I. Pogson. 1983. The control of fatty acid metabolism in liver cells from fed and starved sheep. *Biochem. J.* 214: 553.
- 17 Lopes-Cardozo, M., W. Klazinga, and S. G. van den Bergh. 1978. Evidence for a homogeneous pool of acetyl-CoA in rat liver mitochondria. *Eur. J. Biochem.* 83:635.
- 18 Nosadini, R., H. Datta, A. Hodson, and K.G.M.M. Aberti. 1980. A possible mechanism for the anti-ketogenic action of alanine in the rat. *Biochem. J.* 190: 323.
- 19 Ricks, C. A., and R. M. Cook. 1981. Regulation of volatile fatty acid uptake by mitochondrial acyl-CoA synthetases of bovine liver. *J. Dairy Sci.* 64:2324.
- 20 Robinson, A. M., and D. H. Williamson. 1980. Physiological roles of ketone bodies as substrates and signals in mammalian tissues. *Physiol. Rev.* 60:143.
- 21 Snoswell, A. M., and G. D. Henderson. 1980. Carnitine and metabolism in ruminant animals. Page 191 in *Carnitine biosynthesis, metabolism and functions*. R. A. Frenkel and J. D. McGarry, ed. Academic Press, Inc., New York, NY.
- 22 Steel, R.G.D., and J. H. Torrie. 1980. *Principles and procedures of statistics*. 2nd ed. McGraw-Hill Book Co., New York, NY.
- 23 Thompson, G. E., J. W. Gardner, and A. W. Bell. 1975. The oxygen consumption, fatty acid and glycerol uptake of the liver in fed and fasted sheep during cold exposure. *Q. J. Exp. Physiol.* 60:107.
- 24 Veenhuizen, J. J. 1988. Progressive changes in metabolism of cows during induction of ketosis and treatment of ketosis with intestinal administration of glucose. Ph.D. Diss., Iowa State Univ., Univ. Microfilms, Ann Arbor, MI.
- 25 Veerkamp, J. H., H.T.B. van Moerkerk, J.F.C. Glatz, J.G.E.M. Zuurveld, A.E.M. Jacobs, and A.J.M. Wagenmakers. 1986. $^{14}\text{CO}_2$ Production is no adequate measure of ^{14}C fatty acid oxidation. *Biochem. Med. Metab. Biol.* 35:248.